

# Data Card

## Optimizing RAG Document Retrieval for Agronomic Advice

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|------------------|--------------------------------------------------------|
| Team Name        | RasDashen                                              |
| Selected Project | Optimizing RAG Document Retrieval for Agronomic Advice |
| Challenge        | Challenge 2 — Data Card                                |
| Team Members     | Olumide Adekunle, Misturah, Sabona, Ogwuche Moses      |

## Values-Led Problem Statement (Challenge 1 recap)

Smallholder farmers across many African regions rely on agricultural extension officers for timely, locally relevant advice on crop diseases, pests, soil management, and climate adaptation. Extension services are chronically understaffed relative to the number of farmers who need support, and the written extension material that does exist is scattered across many documents, making it slow for even a knowledgeable officer to locate the single most relevant passage for a farmer's specific question. Our team's values-led aim is to reduce this bottleneck: build a document retrieval system that surfaces the most relevant agronomic extension material for a farmer's question, so that farmers and the extension workers who support them can get trustworthy, source-grounded answers faster. We prioritise **accuracy over fluency** (a retrieval system that returns the wrong document can lead to real crop or livelihood loss), **transparency** (every answer is traceable to a real source document, not a generated claim), and **accessibility** (the system must run on modest, low-cost compute so it can realistically be deployed in low-resource settings).

## 1. Input vs. Output

**Input:** A short natural-language question from a smallholder farmer or extension worker, in text form (e.g. "My maize leaves are turning yellow, what could be wrong?"). Questions are expected to be short (typically one to two sentences), written in plain, non-technical language, and may mix colloquial crop names with formal ones.

**Output:** A ranked list of the top-5 most relevant documents (or document passages) from a fixed knowledge base of agricultural extension material. The model does not generate free-text answers; it retrieves and ranks existing, human-authored source documents. This is a deliberate design choice: it keeps every output traceable to a real, reviewable source rather than a model-generated claim, which matters for a domain where incorrect advice has real-world consequences.

## 2. Classification Task

Not directly applicable — this is a **retrieval/ranking task**, not a classification task. There are no fixed output labels or classes; instead, each candidate document receives a relevance score with respect to a given query, and the top-5 highest-scoring documents are returned. Where useful for analysis, documents are loosely grouped by topical category (e.g. crop diseases, pests, nutrient deficiencies, soil management, climate adaptation, fertiliser advice) purely to support balanced sourcing and evaluation, not as model output classes.

### 3. Labelling Strategy

Relevance labels (which documents are genuinely relevant to a given query, used for local nDCG@5 validation) are sourced from three complementary channels, in order of preference:

- **Existing expert-authored pairings**, where the competition's own labelled training queries (query → known relevant document ids) are used directly.
- **Domain-informed manual review by the team**, for any additional query/document pairs we construct ourselves: at least one team member cross-checks that a document actually answers the query before it is treated as a positive label, rather than relying purely on keyword overlap.
- **AI-assisted candidate generation, human-confirmed**, where an LLM may propose plausible farmer questions for a given extension document to expand our validation set faster, but every AI-proposed pair is reviewed by a team member before being trusted as a label. AI-assisted labels are never treated as ground truth without this human check.

Quality and consistency are checked by having a second team member spot-check a random sample of labelled pairs, and by tracking inter-reviewer agreement informally during team review sessions.

### 4. Data Creation

The document knowledge base draws on **publicly available agricultural extension material**: government and NGO agricultural extension publications, FAO and CGIAR technical guidance, and openly licensed university agricultural-extension course material. Where the competition provides its own corpus of extension documents, that corpus is used as the primary source and is not supplemented with additional scraped content unless the team determines the provided corpus has clear topical gaps. No synthetic (model-generated) documents are added to the knowledge base itself, since the whole point of the system is to point to real, verifiable source material; synthetic data is used only, if at all, to generate additional practice queries for our own internal validation, never as retrievable "documents."

### 5. Web Collection Ethics

Any additional web-sourced material is restricted to content that is explicitly public-domain, openly licensed (e.g. Creative Commons), or published by a government/intergovernmental body (e.g. FAO, national ministries of agriculture) for public extension use. We do not scrape personal data, farmer forum posts, or any content behind logins or paywalls. We record the source URL and licence/attribution for every additional document added, so provenance remains auditable. No personally identifiable information is collected at any stage — the knowledge base consists entirely of institutional/technical extension material, not personal farmer data.

### 6. Bias and Representation

Publicly available agricultural extension material is unevenly distributed: material for widely exported cash crops (e.g. cocoa, coffee) and major staples (maize, rice) is far more abundant than material for regionally important but less "globally visible" crops (e.g. teff, fonio, indigenous vegetables), and English-language material dominates over material in local languages. This risks the system under-serving farmers whose most pressing questions concern under-documented crops or who are more comfortable querying in a local language. To mitigate this, the team will: (1) explicitly audit the topical/crop coverage of the assembled knowledge base

before finalising it, and flag under-represented crop categories rather than silently accepting whatever is easiest to find; (2) prioritise sourcing at least some material on regionally significant but less globally documented crops relevant to our target region; and (3) document known coverage gaps transparently in our final submission rather than presenting the system as universally comprehensive.

## 7. Values and Community Alignment

This dataset design is directly accountable to the values named in our Challenge 1 problem statement: retrieval-only (not generative) outputs preserve **transparency**, since every result traces to a real, human-authored document a farmer or extension worker can independently verify; sourcing only from public-domain/openly-licensed institutional material preserves **trustworthiness and legality**; and the explicit bias audit above is our concrete mechanism for pursuing **equitable access** across crops and regions rather than only the best-documented ones. We remain accountable to the intended community by documenting source provenance and known coverage gaps in this Data Card, so anyone extending this work (including future cohorts) can see exactly what the system does and does not cover, and improve it rather than assume it is complete.

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*Prepared for Challenge 2 (Data Card) submission. Fields in brackets should be completed with team-specific details before submission.*